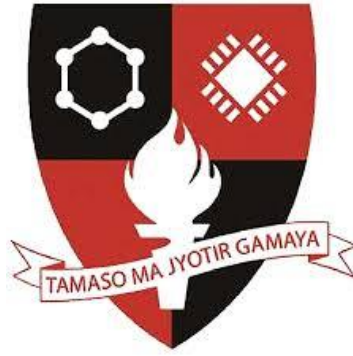




GATE PASS MONITING SYSTEM

SDD

GUIDE

DIVYA S NAIR

GROUP MEMBERS

MEENU V S

PAVITHRA J J

JITHU S SIVAN

RAHUL B

Gate Pass Monitoring System (SDD)

1. Introduction

1.1 Purpose

This Software Design Document (SDD) describes the architecture, design components, data flow, database structure, interfaces, and internal logic of the Gate Pass Monitoring System (GPMS). It provides developers and maintainers with a technical blueprint of the system.

1.2 Scope

The GPMS is a **web-based application** built using:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Django Framework
- **Database:** MySQL

The system automates the process of issuing, approving, scanning, and tracking gate passes within an organization. It includes barcode-based verification and real-time logging.

2. System Architecture

2.1 Architectural Style

The system follows a **Three-Tier Architecture**:

1. **Presentation Layer (Client)**
 - Web browser UI
 - HTML/CSS/JS responsive pages
2. **Application Layer (Server)**
 - Django MVC pattern
 - Business logic & authentication
3. **Data Layer (Database)**
 - MySQL
 - Stores user & gate pass records

2.2 High-Level Architecture Diagram (Conceptual)

Client Browser

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Django Web Server

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MySQL Database

3. System Modules

3.1 Authentication Module

- Handles login & registration
- Encrypts passwords
- Assigns user roles:
 - Advisor
 - HOD
 - Security

3.2 Gate Pass Request Module

- Students submit request form
- Validations applied
- Stored in database with status = “Pending”

3.3 Approval Module

- Advisor reviews & approves/rejects
- HOD performs final approval
- Status updated:
 - Approved
 - Rejected
 - Under Review

3.4 Barcode Module

- Scan uniquely printed barcode
- Linked to student & pass ID
- Appears on ID or web profile

3.5 Verification Module

- Manually contact parents
- Security scans barcode at gate
- System:
 - Validates record
 - Logs exit time
 - Logs entry on return

3.6 Report Module

- Admin/security view movement logs
- Filter by:
 - Date
 - Student
 - Department
 - Status

4. Detailed Design

4.1 Class Design (Conceptual)

Class: Advisor

Attributes:

- user_id
- name
- role
- department
- email
- phone

Class: Gate Pass

Attributes:

- pass_id
- user_id
- reason
- date
- status
- scanned by
- advisor_approval
- created_time

Class: Security

Attributes:

- log_id
- pass_id
- exit_time
- entry_time
- scanned_by

Class:HOD

Attributes:

- log_id
- Department
- report

5. Database Design

5.1 Tables

Table: Advisor

Field	Type
user_id	INT (PK)
name	VARCHAR
email	VARCHAR

Field	Type
role	ENUM
dept	VARCHAR
phone	VARCHAR
password	STAR

Table: gate_pass

Field	Type
pass_id	INT (PK)
user_id	INT (FK)
reason	TEXT
date	DATE
status	ENUM
barcode	VARCHAR
created_at	DATETIME

6. User Interface Design

6.1 Advisor Dashboard

- Request gate pass
- Barcode scanning
- Approve/Reject

6.3 HOD Panel

- Report
- Approval of advisors signup

6.4 Security Panel

- Scan barcode
- Log exit/entry

7. Data Flow Diagrams

7.1 Level-0 DFD

Student ->Barcode -> Apply Pass -> Approval -> Barcode -> Security Scan -> Logs

8. Technology Stack

Component Technology

Frontend HTML, CSS, JS

Backend Django

DB MySQL

Auth Django Auth

Barcode Python libraries / JS

Deployment Server / Localhost

9. Security Design

- Role-based access control
- Password hashing
- Session management
- Input validation
- Protection against:
 - SQL Injection
 - XSS
 - CSRF

10. Performance Considerations

- Optimized database indexing
- Efficient queries
- Caching where applicable
- Supports 500+ users

11. Error Handling

Error	Handling
Invalid login	Show message
Invalid barcode	Access denied
DB fail	Retry
Unauthorized action	Block access

12. Testing Strategy

- Unit testing
- Integration testing
- UI testing
- Security testing
- UAT

13. Future Enhancements

- Mobile App
- Push notifications
- GPS tracking
- Auto-alerts

14. Conclusion

This SDD defines the design and architecture of the Gate Pass Monitoring System. It ensures structured development, maintainability, scalability, and secure operations.